Spatiotemporal Analysis of Coastal Seagrass Dynamics in Florida Bay

Final Project

03/10/2026

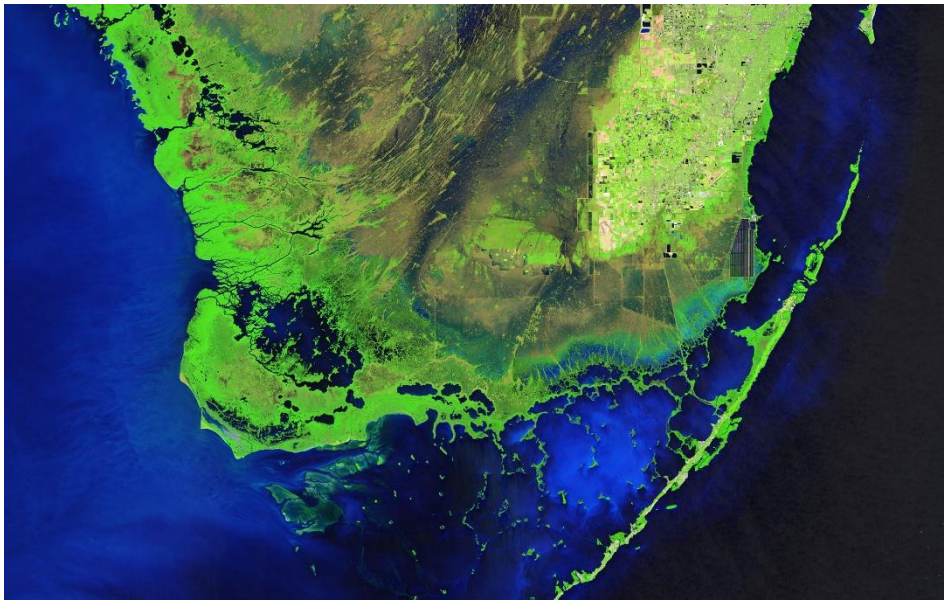

Figure 1. Landsat Vegetation Map (FIU)

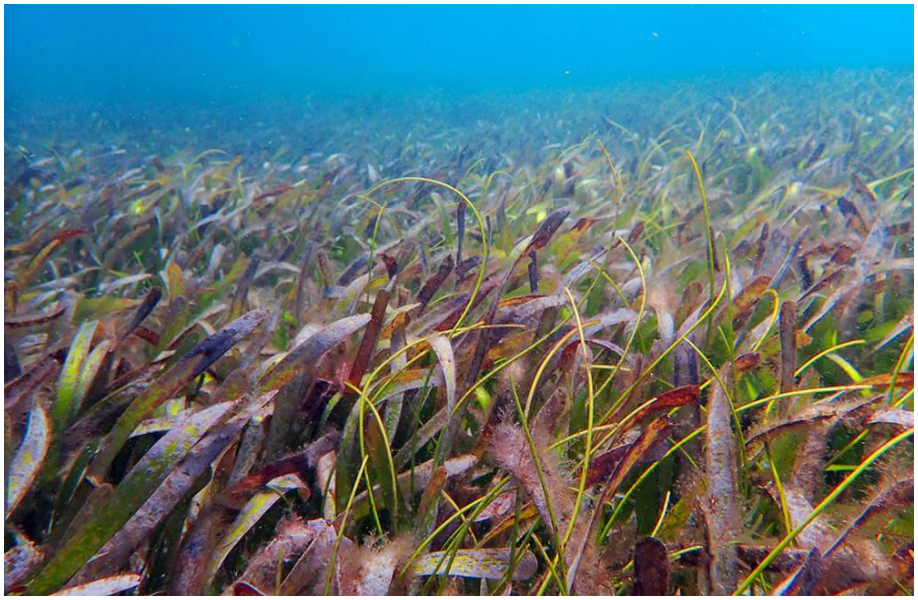

Figure 2. Seagrass meadow (NOAA)

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Acronyms and Abbreviations Used

S2 - Sentinel-2
FL – Florida
WQ – Water Quality
NOAA - National Oceanic and Atmospheric Administration
IEA - Integrated Ecosystem Assessment
GEMM - Groundwater Exchange Monitoring and Modeling plan for Central Florida Bay
FKNMS - Florida Keys National Marine Sanctuary
FIU - Florida International University
SERL –
SECAR - Statewide Ecosystem Assessment of Coastal and Aquatic Resources

Spectral Indices:

NVI - Normalized Vegetation Index
NDVI - Normalized Difference Vegetation Index
EVI - Enhanced Vegetation Index
NDWI - Normalized Difference Water Index
CVI - Coastal Vulnerability Index

Introduction

Florida Bay's seagrass meadows represent one of the largest and most ecologically significant subtropical estuarine ecosystems in North America, serving as a vital habitat for a diverse assemblage of marine species and functioning as a natural system for sediment stabilization, nutrient cycling, and carbon sequestration. Spanning approximately 2,200 square kilometers of shallow coastal water, these meadows form the ecological foundation of South Florida's coastal zone, yet find themselves increasingly threatened by elevated turbidity, water quality degradation, and escalating hyper-salinity. More frequent algal blooms and, since 2015, prolonged drought have intensified these stressors and led to the collapse of approximately 40,000 acres of seagrass meadow in Florida Bay, suggesting a need for the turbidity-corrected, multi-index spectral analysis presented in this study.

Thesis/Problem

This project looks at seagrass in Florida Bay and how it is affected by things like turbidity, salinity, and water quality. Satellite images from Sentinel 2 and NOAA data will be used to see how seagrass changes over time. Python will be used to analyze the data and look for connections between water conditions and where seagrass grows. The goal is to better understand how changes in water affect seagrass in Florida Bay.

Tools & Techniques

This study utilizes two primary data sources: Sentinel-2 multispectral satellite imagery and NOAA oceanic datasets. Sentinel-2 offers publicly accessible satellite imagery at 10-meter spatial resolution with a five-day revisit cycle, allowing for consistent temporal monitoring. From Sentinel-2 band values, four spectral indices are calculated:

The Normalized Difference Vegetation Index (NDVI) quantifies vegetation greenness, indicating seagrass density. The Enhanced Vegetation Index (EVI) improves upon NDVI in optically complex conditions. The Normalized Difference Water Index (NDWI) distinguishes open water from vegetated areas, indicating seagrass distribution. The Turbidity Index quantifies water clarity, identifying where suspended material degrades the accuracy of the vegetation index readings.

Alongside satellite imagery, this study integrates National Oceanic and Atmospheric Administration (NOAA) coastal and oceanic records to characterize the physical environment of Florida Bay. Using multiple indices together provides a clearer picture of seagrass health in Florida Bay; Combining these indices helps improve the analysis and allows the study to better understand how environmental conditions affect seagrass distribution in Florida Bay.

Implementation

The analysis will use Python in JupyterLab, using libraries such as pandas, NumPy, GeoPandas, and working over arrays. A repeatable workflow structure by way of a Project Pythia cookbook through a GitHub repository. The workflow will consist of data access, preprocessing, calculating indices, spatial analysis, and a time-series analysis. An AI LLM system, ChatGPT, will further assist in advancing our coding procedures and analysis.

1. Data Access and Study Area

This study will focus on Florida Bay as the main study area. Satellite images from Sentinel 2 will be used starting from 2015, which is around the time when a large seagrass die-off happened in the bay. Python will be used to process satellite images and calculate the indices NDVI, EVI, NDWI, and turbidity. These indices will help show patterns in seagrass health and water conditions across Florida Bay. Environmental data from NOAA and seagrass monitoring programs will also be used to compare the satellite results with real measurements from the field. This will help us better understand how water conditions are related to changes in seagrass distribution over time.

2. Preprocessing of S2 Imagery

Downloaded satellite imagery from S2 must be preprocessed. The python library Rasterio is mainly used for satellite images because it is raster data. Preprocessing includes:

- **Clipping** images to Study Area: Florida Bay
    - Using a Florida Bay boundary shapefile
    - Using GeoPandas, and checking the coordinate system with ‘bay.to_crs(image.rio.crs)’
    - `rasterio` as `rio`, and `clip` ‘red_clip = red.rio.clip(bay.geometry)’
- **Masking** cloud coverage - Clearing clouds for visibility of the study area
    1. S2 has a Scene Classification Layer (SCL) (example for possible masking)
        - scl = rxr.open_rasterio("SCL.tif")
        - cloud_mask = scl.isin([3,8,9,10]) # cloud categories
        - red_clip = red_clip.where(~cloud_mask)
        - nir_clip = nir_clip.where(~cloud_mask)
    2. By using `rxr.openrasterio` and setting “masked=True”
- **Extracting** the Red, Green, Blue, and near Infrared spectral bands for the indices to be calculated.

Band	Purpose for project
Blue	For EVI
Green	For NDWI
Red	For EVI, NDVI, & Turbidity
NIR (Near Infrared)	For EVI, NDVI, & NDWI

Spectral Bands from S2 imagery will be extracted for index calculations using the Python libraries Rasterio and RioXarray (Xarray for geospatial data), to input each band as an individual raster grid of pixels.

3. Calculations of Indices

Calculations will be executed in Python:.

$$NDVI = (NIR - Red) / (NIR + RED)$$

$$EVI = (2.5 * (NIR - Red) / (NIR+ (6) * Red - (7.5) * Blue + 1))$$

$$NDWI = (Green - NIR) / (Green + NIR)$$

$$Turbidity = Red / Green$$

With the NOAA environmental dataset, we will use various Pandas and Python methods to index by date time, slice and dice, among others, to focus on turbidity and salinity specifically to see if/how it relates or changes the interpretation of the vegetation indices.

4. Spatial Analysis

The temporal spatial analysis will use a shapefile of the Florida Bay to map the calculated indices and determine spatial patterns and relationships within the study area. This will be done using the `.clip()` method within the Python library Xarray, and the Python function `plot` to map the data across Florida Bay. Xarray builds upon NumPy and Pandas. The resulting maps, for NDVI showing seagrass health distribution patterns, NDWI will show water vs vegetation areas in the bay, and a turbidity distribution map.

We will expand further on the spatial analysis by converting the NOAA environmental turbidity data into spatial data, using GeoPandas to compare with the satellite turbidity index using Xarray. Which will provide a measure of whether the calculated satellite turbidity index is accurate for the actual turbidity across Florida Bay.

5. Time-Series Analysis

We will use Pandas `.mean()` for each index and Python `plot` to plot the averages over time in a chart. Because the spectral band data is going to be stored as raster data, we will also use the Pandas `polyfit` module to calculate trend slopes of each pixel over time in the vegetation indices. Thus showing the areas in decline or recovery.

Then we will use the NOAA environmental data, and plot the environmental data for parameters such as turbidity, temperature, and salinity to see how those changed over time. After, we will convert the data to spatial data using x-y coordinates with GeoPandas and Xarray to extract values of the indices from the satellite data and add it to the NOAA data frame for plotting. Then, using the matplotlib library in Python, we will plot the indices with environmental data with respect to time.

Expected Outcomes

Seagrass density and distributions in Florida Bay – a particularly turbid and ecologically dynamic coastal system – tend to decrease when areas of marine habitat experience increased turbidity, hyper-salinity, and/or water quality degradation. This study investigates and quantifies the relationship between these stressors while attempting to improve upon the accuracy of seagrass analysis for Florida Bay's optically complex waters by incorporating the Turbidity Index to correct for water column interference in spectral vegetation outputs. NOAA salinity records will also be considered to investigate any correlation between hyper-salinity events and declines in vegetation index readings. In general, elevated turbidity and hyper-salinity are each expected to show negative correlations with seagrass health across Florida Bay. Results will be presented in a written report, and the analysis pipeline will be hosted on GitHub as a Project Pythia cookbook template.